Replicating Krusell et al. (2017): Gross Worker Flows over the Business Cycle

Macroeconomics with Microeconomic Data

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Abstract

This paper provides an independent Python replication of Krusell et al. (2017), who ask whether a model of individual labour supply with search frictions can account for the cyclical behaviour of gross worker flows across employment (E), unemployment (U) and nonparticipation (N). I reimplement the model using a value–function iteration routine with golden–section savings as well as the Howard improvement for the household problem, and a non–stochastic forward operator for the cross–sectional distribution. These approaches are validated against the published results of KMRS. The replication reproduces the steady–state gross–flow matrix (unemployment rate 0.0678, paper: 0.068) and the business–cycle moments. I discuss the computational choices, how they differ from the original implementation, and two discrepancies worth flagging.

Contents

1 Introduction

Modern models of aggregate labour market dynamics are microfounded on worker flows, allowing them to reconcile micro and macro data. Such models must account not only for the stocks of employed, unemployed, and nonparticipating workers, but also for the gross flows of workers among these states each month. Gross flows count workers moving between states, while net flows are the changes in each stock. Because gross flows are an order of magnitude larger, matching them imposes a far more demanding test than matching the stocks alone.

Krusell et al. (2017), hereafter KMRS, build a parsimonious model to overcome this challenge. Their framework merges two benchmark traditions. The first, rooted in Lucas and Rapping (1969) and Chang and Kim (2006), treats movements between work and non–work as optimal labour–supply responses to prices. The second follows Mortensen and Pissarides (1994), where workers are passive and unemployment is an involuntary friction wedge. Actual labour markets contain both, and whether a transition reflects a choice or a shock matters for the welfare interpretation of joblessness and for policy design. KMRS therefore aim to reproduce the gross flows across all three states.

KMRS's paper rests on two empirical facts. First, although the participation rate barely moves over the business cycle, the gross flows into and out of the labour force are as volatile as those between employment and unemployment. Small net flows mask large gross flows. Second, the flows out of the labour force behave counterintuitively: f_{EN} and f_{UN} are procyclical, rising in good times even as participation itself is rising.

The authors calibrate the model to the average gross–flow matrix, then shock the frictions alone, with prices held constant. This is sufficient to reproduce the cyclical patterns above. The central result concerns participation, which the model makes only weakly procyclical. Two forces are at play. On one side, a wealth effect: in good times, higher income reduces the desire to work. On the other, the intertemporal substitution effect: even at a fixed wage, offers arrive faster in good times, and each offer is an opportunity to climb the job ladder, so employment becomes more valuable. The two nearly offset each other, hence participation barely moves over the cycle.

This paper presents a computational replication of KMRS in Python. The official replication package published by the AER builds the empirical flows from CPS and SIPP microdata and solves the model. Instead, I reimplement the model independently, treating the authors' results as the validation targets. The replication package was consulted only for implementation conventions that the paper leaves unstated. My version speeds up the computation in two ways: an accelerated value–function solver, as discussed in Goensch (2025), and two convenient features of the search–cost shock. Neither change the conclusions. The replication therefore closely matches the original results.

2 Model

The economy is populated by a continuum of ex–ante identical, infinitely–lived households subject to idiosyncratic productivity, taste and match–quality shocks, in the presence of incomplete markets and search frictions. There is no explicit firm side and markets do not clear every period. The analysis is partial equilibrium, though prices are pinned down by a background general equilibrium (Section 2.7). KMRS study only the labour–supply side, but show that their framework is consistent with a general–equilibrium economy of two islands: a work island (employment) and a leisure island (nonemployment).

2.1 Preferences

A household orders consumption and labour–market activity according to:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \big[\ln c_t - \alpha\, e_t - \gamma_t\, s_t \big], \qquad 0 < \beta < 1 \tag{1}$$

where $c_t \geq 0$ is consumption, $e_t \in \{0,1\}$ indicates employment, and $s_t \in \{0,1\}$ indicates active job search. The parameter $\alpha > 0$ is the disutility of work and $\gamma_t > 0$ the disutility of active search. Log utility implies a unit intertemporal elasticity of substitution. The two discrete variables (e and s) account for extensive–margin labour supply (there is no intensive margin in the model).

2.2 Idiosyncratic shocks

Three idiosyncratic shocks drive individual behaviour.

(i) **Market productivity** z, persistent and AR(1) in logs:

$$\ln z_{t+1} = \rho_z \ln z_t + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2) \tag{2}$$

(ii) **Search–cost** γ, transitory and i.i.d., drawn each period from a three–point distribution with equal probability on $\{\bar{\gamma} - \varepsilon_\gamma,\ \bar{\gamma},\ \bar{\gamma} + \varepsilon_\gamma\}$. This taste shock generates the large $U \leftrightarrow N$ flows observed in the data.

(iii) **Match quality** q, permanent within a match, drawn from a log–normal distribution with $\ln q \sim \mathcal{N}(0, \sigma_q^2)$ when an employment opportunity arrives and observed at that point. Match quality, together with on–the–job search, creates a job ladder.

An employed worker's labour earnings are the product $w\,z\,q$, where w is the aggregate wage per efficiency unit.

2.3 Frictions and unemployment insurance

Labour–market conditions are summarised by four friction parameters. λ_u is the probability that a non–employed individual who searches actively receives an employment opportunity, and λ_n the same under passive search, where $\lambda_n < \lambda_u$. λ_e is the probability that an employed individual receives an outside offer (on–the–job search), and σ is the exogenous separation rate. A worker who has just lost their job receives a fresh offer with probability λ_u, the same rate as an active searcher.

The unemployment–insurance (UI) system ties benefits to involuntary job loss and active search. Eligibility (indicator $I^B \in \{0, 1\}$) requires a previous involuntary separation. Since UI benefits have finite duration, eligibility expires each period with probability μ.

Benefits are collected only while searching actively and equal:

$$b(z) = \min\{\, b_0\, z,\ \bar{b}\,\} \tag{3}$$

indexed to current productivity and capped at $\bar{b}$.

2.4 Budget constraint

Markets are incomplete. There is only one asset $a \geq 0$ with a rate of return r. There is a proportional tax τ on labour income and a lump–sum transfer T. The period budget constraint is:

$$c_t + a_{t+1} = (1+r)\ a_t + (1-\tau)\ w\ z_t\ q_t\ e_t + (1-e_t)\ I^B\ s_t(1-\tau)\ b(z_t) + T \tag{4}$$

2.5 Timing and the recursive problem

At the start of a period the household observes $(z, \gamma, I^B, \Lambda)$, where the vector $\Lambda \equiv (w, r, \lambda_u, \lambda_n, \lambda_e, \sigma)$ collects aggregate variables. Decisions are made after all shocks take place. The state of an individual is $(a, z, \gamma, I^B, \Lambda)$, augmented by match quality q for those holding a job or an offer. Define the value of being jobless (no offer in hand) as the maximum of active versus passive search:

$$J(a,z,\gamma,I^B,\Lambda) = \max\left\{\, U(a,z,\gamma,I^B,\Lambda),\; N(a,z,\gamma,I^B,\Lambda)\,\right\} \tag{5}$$

and the value of holding an offer of quality q as the maximum of either accepting (W) or rejecting it (J):

$$V(a,z,q,\gamma,I^B,\Lambda) = \max\left\{\, W(a,z,q,I^B,\Lambda),\; J(a,z,\gamma,I^B,\Lambda)\,\right\} \tag{6}$$

Economically, Equation (5) is the participation decision: paying the utility cost γ is an investment that buys a higher offer–arrival rate and, if eligible, UI receipt. Equation (6) is a reservation–quality decision. The employed value solves:

$$\begin{aligned} W(a,z,q,I^B,\Lambda) = \max_{c\geq 0,\, a'\geq 0} \Big\{ \ln c - \alpha + \beta\,\mathbb{E}\Big[&(1-\sigma-\lambda_e)\,V(a',z',q,\gamma',0,\Lambda') \\ &+ \lambda_e\,V(a',z',\max\{q,q'\},\gamma',0,\Lambda') \\ &+ \sigma(1-\lambda_u)\,J(a',z',\gamma',1,\Lambda') \\ &+ \sigma\lambda_u\,V(a',z',q',\gamma',1,\Lambda')\Big]\Big\} \end{aligned} \tag{7}$$

subject to the budget constraint in Equation (4) with $e = 1$. The four continuation terms are mutually exclusive. With probability $1-\sigma-\lambda_e$ the worker keeps the job and does not get an offer. With λ_e, he or she keeps the job and draws an outside offer, accepting the one with higher match quality ($\max\{q,q'\}$ i.e. the job ladder). $\sigma\lambda_u$, worker separates, but immediately draws a new offer. Lastly, with $\sigma(1-\lambda_u)$, there is separation and no offer. Separation confers UI eligibility ($I^{B\prime} = 1$), while staying employed does not ($I^{B\prime} = 0$).

The active– and passive–search values are:

$$\begin{aligned} U(a,z,\gamma,I^B,\Lambda) = \max_{c,a'\geq 0} \Big\{ \ln c - \gamma + \beta\,\mathbb{E}\big[&\lambda_u\,V(a',z',q',\gamma',I^{B\prime},\Lambda') \\ &+ (1-\lambda_u)\,J(a',z',\gamma',I^{B\prime},\Lambda')\big]\Big\} \end{aligned} \tag{8}$$

$$\begin{aligned} N(a,z,\gamma,I^B,\Lambda) = \max_{c,a'\geq 0} \Big\{ \ln c + \beta\,\mathbb{E}\big[&\lambda_n\,V(a',z',q',\gamma',I^{B\prime},\Lambda') \\ &+ (1-\lambda_n)\,J(a',z',\gamma',I^{B\prime},\Lambda')\big]\Big\} \end{aligned} \tag{9}$$

subject to the budget constraint with $e = 0$ and, $s = 1$ (UI received if eligible) and $s = 0$ (no search cost, no UI). An eligible individual remains eligible next period ($I^{B\prime} = 1$) with probability $1-\mu$.

2.6 Mapping to labour–market states

The model maps directly to the data. An individual who works in period t is employed, whereas one who is not employed but searches actively (chooses U) is unemployed. All those who are not employed and do not search actively (choose N) are out of the labour force. The decision rules assign each individual to E, U or N. Gross flows are the transitions between them as individual circumstances change.

2.7 Aggregation and stationary equilibrium

With a continuum of such households and fixed aggregate conditions there is an invariant joint distribution over individual states, inducing invariant distributions over the three stocks and over the gross flows. The analysis is partial equilibrium, that is, prices w and r are exogenous to the household and held constant. Following KMRS, the constant steady–state prices are nonetheless disciplined to be consistent with a background Cobb–Douglas general equilibrium. With capital share α_K and aggregate efficiency units L:

$$w = (1 - \alpha_K)\,(K/L)^{\alpha_K}, \qquad r = \alpha_K\,(K/L)^{\alpha_K - 1} - \delta \tag{10}$$

and the lump–sum transfer T balances the government budget ($T = \text{taxes} - \text{UI}$). The capital–labour ratio K/L equals aggregate household wealth divided by aggregate efficiency units.

2.8 Aggregate shocks

Over the cycle the only source of aggregate variation is the frictions. The wage w and the interest rate r are held constant. Aggregate conditions follow a symmetric two–state Markov chain (Good/Bad) in which each state persists with probability ρ. The good state has high arrival rates and a low separation rate:

$$\lambda_u^G = \lambda_u^* + \varepsilon^\lambda, \quad \lambda_u^B = \lambda_u^* - \varepsilon^\lambda, \quad \sigma^G = \sigma^* - \varepsilon^\sigma, \quad \sigma^B = \sigma^* + \varepsilon^\sigma \tag{11}$$

with λ_e and λ_n moving so as to preserve constant ratios to λ_u. The amplitudes are chosen so that the simulated volatilities of f_{UE} and f_{EU} match their empirical counterparts.

2.9 Calibration

The period is one month. Several parameters are set outside the model, namely the idiosyncratic process from estimates of the persistence and standard deviation of individual wages (annual $\rho_z = 0.955$, $\sigma_\varepsilon = 0.20$, converted to monthly), the tax rate, the

UI duration and replacement, and the prices via Equation (10) with $\alpha_K = 0.30$. The remaining parameters $\{\alpha, \sigma, \lambda_u, \lambda_n, \lambda_e, \sigma_q, \varepsilon_\gamma\}$ are chosen jointly so that the steady state matches a set of average targets: the labour–force participation rate (0.66), the unemployment rate (0.068), the $E \to U$ rate (0.014), the $N \to E$ rate (0.022), the job–to–job rate (0.022), the average wage gain on a job–to–job move (0.033), and the magnitude of the $U \leftrightarrow N$ flows. Although jointly determined, each parameter has a natural primary target (last column of Table 1), which keeps the calibration transparent. Table 1 reports the benchmark values, which reproduce Table 4 of KMRS.

Parameter	Symbol	Value	Target / Source
Discount factor	β	0.99465	background GE, $1+r=1.00327$
Persistence of $\ln z$	ρ_z	0.996	annual 0.955
Innovation s.d. of $\ln z$	σ_ε	0.096	annual 0.20
UI expiry	μ	1/6	6–month duration
Disutility of work	α	0.485	lfpr $= 0.66$
Mean search cost	$\bar{\gamma}$	0.042	$= \alpha \cdot 3.5/40$ (ATUS)
Active arrival	λ_u	0.278	unemployment rate $= 0.068$
Passive arrival	λ_n	0.182	$N \to E$ rate $= 0.022$
Separation rate	σ	0.0178	$E \to U$ rate $= 0.014$
On–the–job arrival	λ_e	0.121	job–to–job rate $= 0.022$
Match–quality s.d.	σ_q	0.034	J2J wage gain $= 0.033$
Search–cost spread	ε_γ	0.030	$U \leftrightarrow N$ flows
Tax rate	τ	0.30	effective labour tax
Capital share	α_K	0.30	NIPA
Aggregate shock amplitudes	$\varepsilon^\lambda, \varepsilon^\sigma$	0.0662, 0.00239	s.d. of f_{UE}, f_{EU}
Aggregate persistence	ρ	0.983	business–cycle persistence

Table 1: Calibration (monthly). The replication uses $\lambda_u = 0.282$ from the authors' code (while the paper's Table 4 prints 0.278), from which λ_e and λ_n are derived.

3 Computational Methods

Table 2 summarises the computational choices, and how they relate to the original authors' implementation. This section describes each step.

3.1 Discretisation of the shocks

The persistent shock z follows the AR(1) in Equation (2). Following Tauchen (1986), I approximate it by a Markov chain with $n_z = 20$ states. The grid is evenly spaced in logs and spans ± 2 unconditional standard deviations, and the transition probabilities come from integrating the conditional normal over midpoint bins. The annual persistence and innovation standard deviation are converted to monthly values exactly as in the authors' Fortran code.[1] Match quality is discretised on $n_q = 7$ points over $\pm 2\sigma_q$ with normal–bin weights, and the search cost on $n_\gamma = 3$ equally weighted points. The aggregate chain has $n_z^{\text{agg}} = 2$ states.

3.2 Asset grids

The model uses two asset grids. The household problem is solved on a coarse grid of $n_a = 48$ points, evenly spaced in $\log(a+2)$ over $[0, 1440]$ so that points cluster near the borrowing constraint, where the saving policy has most curvature. The cross–sectional distribution uses a finer grid with $n_a^{\text{fine}} = 1000$ points, evenly spaced in levels, which keeps its resolution uniform across the wealth distribution.

3.3 Solving the household problem

The model has five value functions: the employed value W, and the active– and passive–search values U and N at each UI–eligibility level. All are solved using value–function iteration (VFI). At each grid node the continuation value is already integrated over next period's $(z', q', \gamma', I^{B\prime})$ using the discretised processes. It is interpolated linearly in a', and the savings choice is achieved via a vectorised golden–section search. To accelerate convergence, I wrap the iteration in a Howard policy–improvement step, as seen in Goensch (2025), which converges to the same fixed point as the "pure" value iteration used by KMRS.[2]

3.4 The stationary distribution

Given the policy functions, I compute the invariant distribution with the non–stochastic forward operator of Young (2010), which KMRS also use. Advancing the distribution one period sends each household to its next–period state. Assets and productivity update mechanically. The saving policy moves assets from a to a' (split between the two bracketing grid nodes), and productivity follows the Markov chain, which spreads the mass over next period's z. The labour–market state updates through the period's random events. An employed household may keep its job, draw an outside offer, or separate, while a non–employed one may or may not receive an offer, and UI eligibility may lapse. Each realisation of these events is a branch, reached with a known probability. Within each branch, the household chooses whether to work, search actively, or stay out, taking whichever is best given a freshly drawn search cost γ' and, where an

[1] $\rho_z = \rho_{\text{ann}}^{1/12}$: the monthly innovation standard deviation is chosen so that the implied twelve–month aggregate reproduces the annual standard deviation, yielding $\sigma_\varepsilon = 0.0957$.

[2] This acceleration does not change the fixed point. When the model is solved at the authors' converged prices, the steady–state gross flows still match their results.

offer arrived, its match quality q'. The branch it passed through fixes its UI eligibility. Gross flows are recorded in the same pass, and the operator is iterated to a tolerance of 10^{-9}.

Because the value functions are fixed while the distribution converges, these labour–market choices never change. I therefore compute them once, up front, so that each forward iteration boils down to a few inexpensive array operations.[3]

3.5 Calibrating the constant prices (background general equilibrium)

Because the model is partial equilibrium, prices w and r are fixed inputs. However, their level must be pinned down. Following KMRS, I require the steady state to be consistent with a background general equilibrium: a damped fixed point on the capital–labour ratio K/L, the average earnings level (which enters the UI cap), and the transfer T. Each candidate K/L implies prices through Equation (10). Solving the household problem and the stationary distribution yields new values for all three objects, which are then updated with dampening until they converge. The calibration is run once, for the steady state. The constant prices it produces are used unchanged thereafter, including over the business cycle. Algorithm 1 states the procedure and Figure 1 depicts the full pipeline.

Algorithm 1 Background–GE calibration of the constant steady–state prices

Initialise K/L, average earnings $\bar{w}$, transfer T, and tolerances.
for $s = 1, 2, \ldots$ **do**
  Prices: w, r from K/L via Equation (10), UI cap from $\bar{w}$.
  Household: solve the five value functions by VFI with Howard steps.
  Distribution: iterate the forward operator (Section 3.4) to the invariant distribution.
  Aggregates: $K/L^* =$ wealth/efficiency units, $\bar{w}^* = \mathbb{E}[zq \mid E]$, $T^* =$ taxes$-$UI.
  Converged? if $|K/L^* - K/L|$, $|\bar{w}^* - \bar{w}|$, $|T^* - T|$ small, stop.
  Update with dampening: $K/L \leftarrow 0.1\, K/L^* + 0.9\, K/L$, and $\bar{w}$, T with weight 0.5.
end for

[3]The reorganisation is exact, not an approximation. It keeps the price–calibration and business–cycle loops practical. A full run takes about twelve minutes.

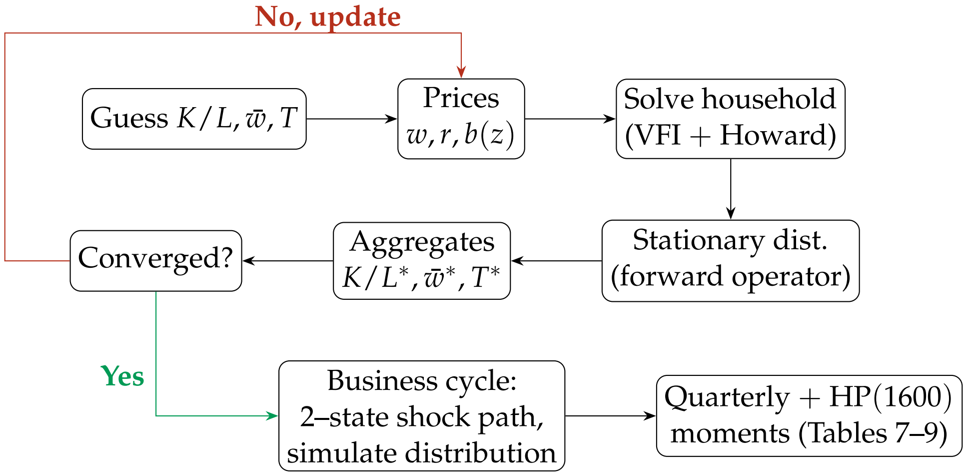

Figure 1: Solution pipeline. The top loop calibrates the constant prices to a background general equilibrium (the model itself is partial equilibrium). The converged distribution and value functions then seed the business–cycle simulation (bottom), which uses fixed prices.

3.6 Business–cycle simulation

The cyclical experiment subjects the model to the two–state shock in Equation (11). The value functions are first re–solved over both aggregate states (the continuation uses the next state's frictions and values, combined with the aggregate transition matrix, as in the authors' business–cycle code). A shock path of 5,000 months is drawn from the symmetric chain (the first period is the good state), and the cross–sectional distribution is propagated along it. The simulation is not a Monte Carlo panel. The full distribution is propagated deterministically. The timing follows the model. The saving policy comes from the current aggregate state, while the events and discrete choices use next–state conditions. Each month the gross flows, stocks and output $Y = w\,L/(1-\alpha_K)$ are recorded. The first 1,000 months are discarded. The series are then aggregated to quarterly frequency, using three–month means for rates and sums for output, then logged and HP–filtered with smoothing parameter 1600. The filter is implemented in–house as a sparse $(I + \lambda D'D)$ solve, the same operation as in the original model.

Component	This replication	KMRS (2017)
Language	Python	Fortran + MATLAB stats
Household problem	VFI, golden section, Howard improv.	VFI, golden section
Shock discretisation	Tauchen $n_z = 20$, $n_q = 7$, $n_\gamma = 3$	identical
Asset grids	48 (log) solve / 1000 (linear) dist.	identical
Distribution	Young (2010), pre–built operator	Young (2010), explicit loops
Search–cost state	analytic ($U = \hat{U} - \gamma$)	explicit γ dimension
General equilibrium	damped fixed point on $K/L, \bar{w}, T$	damped fixed point
Business cycle	deterministic distribution along path	identical
Statistics	in–house HP(1600), quarterly	`hpfilter.m`, quarterly

Table 2: Computational methods: this replication versus the original.

4 Results

I validate the replication against the model columns of the published KMRS tables.

4.1 Average gross flows (Table 5) and the calibrated prices

Table 3 reports the average monthly transition matrix in the calibrated steady state. Every flow lies inside the data 95% confidence intervals reported by KMRS, and all of them reproduce the authors' model column to rounding. The calibration also re–derives $K/L = 129.53$, average earnings $\bar{w} = 2.477$ and transfer $T = 1.357$, against 129.51, 2.477 and 1.357 in the authors' code.[4]

Each month roughly 22% of the unemployed move into employment and 13% leave the labour force, while over 4% of nonparticipants enter it. The flows through N are of the same order as those between E and U, yet the stocks are constant: the movements exactly offset. Matching the $U \leftrightarrow N$ cells, which KMRS note no previous structural model had fit, rests on two features. The participation choice moves marginal workers across the U/N boundary, while frictions ($\lambda_u < 1$) keep active searchers unemployed involuntarily.

	Model			Paper			Data		
From \ To	E	U	N	E	U	N	E	U	N
E	0.972	0.014	0.014	0.972	0.014	0.014	0.972	0.014	0.014
U	0.219	0.652	0.130	0.219	0.652	0.130	0.228	0.637	0.135
N	0.022	0.020	0.958	0.022	0.020	0.958	0.022	0.021	0.957

Table 3: Average gross worker flows in the steady state (monthly transition rates). "Model" is this replication. "Paper" is the KMRS model. "Data" is the Abowd–Zellner–adjusted CPS. $E \to E$ aggregates stayers and job–to–job movers.

4.2 Flows by wealth (Table 6)

The previous section showed that the model reproduces the aggregate gross flows in the data. The model also has predictions for the pattern of gross flows by wealth, i.e. conditional on one of the state variables. Table 4 reports flow rates by wealth quintile relative to the aggregate. These rates reproduce the model column of KMRS's Table 6. The qualitative patterns KMRS emphasise are all present, each reflecting a wealth or a job–ladder effect. The $E \to U$ rate falls with wealth because a separated worker with assets can afford to withdraw rather than search. Exits to nonparticipation instead fall to a minimum in the middle of the distribution and rise again at the top. In the model

[4] The results above use the self–calibrated prices. A fixed–price check (`python main.py -quick`) instead imposes the authors' converged price constants. The quintile rates in Table 4 come from it.

the asset–poor leave when low productivity makes work barely worthwhile, and the asset–rich because leisure is a normal good. The steep job–to–job gradient (1.68 in the bottom quintile) reflects their position on the ladder. Wealthier workers have had time to climb into better matches, so fewer arriving offers beat their current match. As in the original, the model's gradients are starker than the SIPP data (a bottom–quintile $E \to N$ ratio of 1.15 in the data against 4.54 in the model), and the wealth gradients of the $U \to E$ and $U \to N$ rates do not match the data qualitatively. KMRS deem the $U \to E$ departure a direct consequence of the parsimony of their model, and in particular of assuming homogeneous arrival rates.

Flow	$Q1$	$Q2$	$Q3$	$Q4$	$Q5$
$E \to U$	1.13	1.05	0.97	0.97	0.96
$E \to N$	4.54	1.24	0.54	0.63	0.72
$U \to E$	1.05	0.92	1.02	1.03	1.02
$U \to N$	1.54	1.04	0.81	0.87	0.80
$N \to E$	1.16	1.33	0.89	0.70	0.73
$N \to U$	0.73	1.87	1.11	0.88	0.83
Job–to–job	1.68	1.01	0.92	0.96	0.95

Table 4: Flow rates by wealth quintile relative to the aggregate (replication model).

4.3 Cyclical behaviour of stocks (Table 7)

Table 5 reports the business–cycle statistics of the three stocks. The model reproduces the KMRS values closely, and with them the headline result of the paper: the participation rate is weakly procyclical (corr $= 0.39$, paper 0.37) and an order of magnitude less volatile than the unemployment rate, while employment is strongly procyclical.

The economics behind this muted response reflects two opposing forces acting on the same marginal worker. In good times the return to being in the labour market is high. Offers arrive faster, jobs last longer and, most importantly, outside offers reach the employed more often, so individual wages grow faster along the job ladder even though the wage per efficiency unit never moves. On its own this pulls workers in. But the same improvement raises expected lifetime income, and since leisure is a normal good the resulting wealth effect pushes the now richer households out. In steady–state comparisons the wealth effect dominates and participation is countercyclical (Krusell et al., 2010). In the calibrated model the two forces nearly cancel out. Participation volatility (0.0016) is about 60% of that in the data (0.0026), with the correct sign and persistence. The model accounts for more than half of observed participation movements with frictions as the only source of shocks.[5] The perfect $\mathrm{corr}(u, Y) = -0.99$ (data -0.84) reflects

[5] KMRS establish that the procyclical side comes from on–the–job search: removing it turns participation strongly countercyclical. That experiment is not re–run here.

the single–shock structure: with one aggregate state driving all frictions, model aggregates comove almost perfectly, whereas the data mix several impulses and measurement error.

	std			corr$(\cdot, Y)$			corr$(\cdot, \cdot_{-1})$		
	Model	Paper	Data	Model	Paper	Data	Model	Paper	Data
u	0.128	0.121	0.117	-0.99	-0.99	-0.84	0.88	0.87	0.93
lfpr	0.0016	0.0015	0.0026	0.39	0.37	0.21	0.74	0.71	0.69
E	0.0102	0.0096	0.0099	0.996	0.995	0.83	0.89	0.89	0.92

Table 5: Cyclical properties of stocks (quarterly, logged, HP–filtered, $\lambda = 1600$). "Model" is this replication. "Paper" is KMRS. "Data" comes from the CPS.

4.4 Cyclical behaviour of flows (Tables 8 and 9)

Table 6 reports the cyclical statistics of the six gross flows and the job–to–job rate. The model reproduces every sign and matches the magnitudes closely. The largest discrepancy is 0.05, in the autocorrelation of f_{UN}. It also captures the first key fact of the paper, that in the data the flows into and out of the labour force are as volatile as the $E \leftrightarrow U$ flows. Figure 2 displays the flow cyclicalities and volatilities against the paper's model results.

The countercyclical f_{EU} and the procyclical f_{UE} follow mechanically from the shocks to σ and λ_u, whose amplitudes were chosen to match those two volatilities, so the remaining flows are the genuine predictions. The most striking are the two KMRS call counterintuitive: both rates out of participation rise in good times, f_{EN} with a correlation of $+0.24$ and f_{UN} of $+0.48$, even though the stock of nonparticipants is countercyclical. In KMRS's world this is a composition effect. In booms, job losers exit unemployment quickly, so recent entrants from nonparticipation make up more of the unemployed pool. These are weakly attached workers close to the U/N boundary, who flow back to N at high rates. In recessions, the pool is filled with displaced workers who search hard, and the average exit rate to N falls. Discouragement would predict the opposite, a higher exit rate to N in slumps. The same boundary logic underlies the procyclical f_{EN}: marginal workers drawn into employment in good times are those most likely to flow back out. This replication reproduces these patterns. The compositional shift behind them is what KMRS document in cross–sectional data (their Table 10). On magnitudes, the model understates the volatility of f_{UN} relative to the Abowd–Zellner–adjusted data (0.029 against 0.106), although the alternative deNUNified correction (0.066) is considerably closer. Authors argue the discrepancy should be discounted accordingly.

The job–to–job rate (Table 6, last row) is procyclical (corr $= 0.55$) and volatile (std $=$ 0.103). Procyclical job–to–job flows are also part of the mechanism behind Table 5: faster ladder–climbing in booms generates procyclical individual wage growth at a constant wage per efficiency unit, which is precisely the aforementioned intertemporal–substitution channel.

	std		$\mathrm{corr}(\cdot, Y)$		$\mathrm{corr}(\cdot, \cdot_{-1})$	
Flow	Model	Paper	Model	Paper	Model	Paper
f_{EU}	0.094	0.089	-0.79	-0.79	0.78	0.76
f_{EN}	0.057	0.057	0.24	0.21	0.25	0.21
f_{UE}	0.092	0.088	0.70	0.69	0.73	0.70
f_{UN}	0.029	0.029	0.48	0.47	0.39	0.34
f_{NE}	0.053	0.051	0.58	0.57	0.68	0.66
f_{NU}	0.081	0.076	-0.96	-0.96	0.88	0.87
Job–to–job	0.103	0.098	0.55	0.54	0.74	0.72

Table 6: Cyclical properties of gross flows and the job–to–job rate. "Model" is this replication. "Paper" is the KMRS model (Tables 8C and 9).

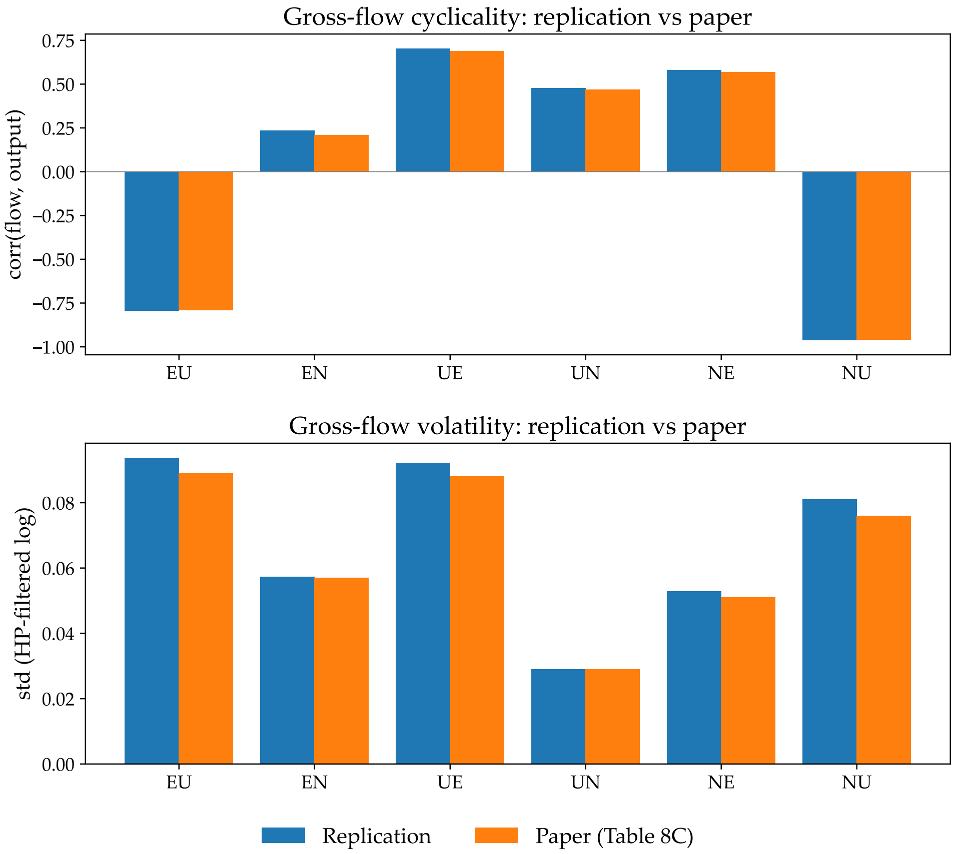

Figure 2: Cyclicality (top) and volatility (bottom) of the six gross flows.

4.5 Variance decomposition (Table 11)

Following Elsby et al. (2015), changes in the unemployment rate can be attributed to the flows between each pair of labour–market states. The economic question is how much of unemployment's cyclical movement runs through the participation margin rather than through job finding and job loss. In both KMRS's model and the adjusted data the answer is roughly 30%, which is the basis for the paper's conclusion that participation flows are a quantitatively important driver of unemployment fluctuations. My implementation builds the flow–steady–state unemployment rate each period, then attributes its cyclical variance to the three pairs by a counterfactual–covariance method. It reproduces the coarse structure. Flows involving unemployment dominate (jointly $\sim 98\%$), and the $E \leftrightarrow N$ contribution is negligible. However, it splits the total differently (85, 13, 2 against the published 74, 31, -4). The division between $U \leftrightarrow E$ and $U \leftrightarrow N$ is sensitive to the decomposition method, and the exact Elsby et al. (2015) implementation used by KMRS is carried out in an external Excel spreadsheet rather than in the code. The replication therefore supports the qualitative claim that participation flows contribute to unemployment fluctuations. The $\sim 30\%$ share itself is not independently verified.

4.6 Discussion of discrepancies

The replication reproduces the authors' results with high precision. Two small discrepancies worth mentioning remain.

(i) Business–cycle volatilities: the model standard deviations are uniformly about 5% larger than KMRS's. This is a sampling difference, not a model error. Fortran and Python generate different random shock paths even under the same seed, so the two simulations average slightly different samples. The correlations and autocorrelations, which are far less sensitive to the path, still match to ~ 0.02, and the gap is uniform across all series.

(ii) Variance decomposition: as noted, the $U \leftrightarrow E$ versus $U \leftrightarrow N$ split depends on the exact decomposition method, which is implemented in an external Excel spreadsheet. The qualitative result is nevertheless reproduced.

5 Conclusion

This paper provides an independent Python replication of Krusell et al. (2017). The steady–state gross–flow matrix and the wealth–conditioned flows match the paper's results (Tables 3–4), and the background GE price calibration independently re–derives the constant prices. Subjecting the model to shocks to frictions alone, with the wage per efficiency unit held constant, reproduces the cyclical behaviour of the three stocks and the six gross flows (Tables 5–6), including the paper's central result that the participation rate is weakly procyclical and an order of magnitude less volatile than unemployment, and the counterintuitive procyclicality of f_{EN} and countercyclicality of f_{NU}.

The replication shows that these results are well established properties of the model. An incomplete–markets economy in which participation is chosen and jobs arrive through frictions can be calibrated to the average gross–flow matrix and then, fed friction shocks of empirically plausible size, with wages held constant, deliver the joint cyclical behaviour of the three stocks and the six flows. In the language of Krusell et al. (2010), joblessness in this economy reflects both choice and chance. Involuntary separations coexist with an operative labour–supply margin, and the two forces on participation nearly offset each other, keeping it stable over the cycle even as large gross flows continue underneath. The framework thus combines the Lucas and Rapping (1969) and Mortensen and Pissarides (1994) traditions, and provides a natural setting for studying policies, such as unemployment insurance, whose incentives act on the search and participation margins.

This implementation departs from the original in two computational aspects. The value–function iteration is accelerated with a Howard policy–improvement step, and two exact features of the search–cost shock (it enters additively and is redrawn each period) remove it from the savings problem and from the cross–sectional state. Both choices leave the fixed point unchanged, as the flow match at the authors' converged prices confirms. The two remaining discrepancies, a shock–path generator difference, and an external spreadsheet for the variance decomposition, are documented and well understood.

Disclaimer on AI usage: *I used a large language model (Anthropic's Claude) as an assistant and referee in this project. It helped me interpret the authors' original code and check my work against theirs. It assisted in writing and reviewing the Python implementation. Regarding this term paper, it helped improve sentence clarity, check alignment with the code and the original paper, and spot typos or any grammar mistakes. The code, modelling choices, calibration, results, and paper are my own. I therefore take full responsibility for any errors.*

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